Paper 36: ξ_{eff} is Set by Wave Radius, Not Diffusion

Testing the DIFFUSE $\to \xi \to N_{\text{crit}}$ Prediction from Papers 3 and 35;
Correction to the Inter-Module Coupling Bandwidth Formula

Gabriel Aubin-Moreau

March 2026

Abstract

Paper 35 derived the relay coupling threshold $N_{\text{crit}} = L/(4\xi)$, where ξ is the spatial correlation length and $L = 40$ sites is the lattice span. Combining with Paper 3's scaling law $\xi \propto \sqrt{\kappa/\nu} \propto \sqrt{D}$ (with $D = \texttt{DIFFUSE}$) yields the prediction $N_{\text{crit}} \propto 1/\sqrt{D}$: a 16× variation in D should produce a 4× variation in N_{crit}, spanning $N_{\text{crit}} \in \{2, 4, 8\}$. We test this prediction by running the relay gain experiment (Paper 35, Exp B) at $D \in \{0.005, 0.020, 0.080\}$ (220 total simulations). The prediction fails: $N_{\text{crit}} \approx 5$ for *all three* DIFFUSE values. The correct formula is $N_{\text{crit}} = L/(4r_{\text{wave}})$ where $r_{\text{wave}} = 2$ sites is the wave footprint radius — a fixed system property independent of D. The effective correlation length $\xi_{\text{eff}} \approx \max(\xi_D, r_{\text{wave}})$, saturating at r_{wave} both when $\xi_D < r_{\text{wave}}$ (diffusion too slow; wave-dominated) and when $\xi_D > r_{\text{wave}}$ (zone-wave reinforcement prevents diffusion from expanding the boundary width beyond the wave footprint). Paper 3's $\xi \propto \sqrt{D}$ law governs the diffusion-limited regime in a uniform medium; at zone boundaries under active wave-driven reinforcement, r_{wave} is the effective ceiling. This is a correction to Papers 3 and 35: the inter-module coupling bandwidth is set by the wave radius, not the diffusion length.

1 Introduction and Motivation

Paper 35 established two key results for geographic relay coupling:

1. The relay gain G peaks at optimal wave rate WR ≈ 4.8 (wave flux $\Phi = \text{WR} \cdot r_{\text{wave}}/T_{\text{dur}} \approx 0.64$).
2. G collapses when zone width $z_w < 4\xi_{\text{eff}}$, establishing $N_{\text{crit}} = L/(4\xi_{\text{eff}})$ with $\xi_{\text{eff}} \approx 2.5$ sites at standard parameters.

The correlation length $\xi_{\text{eff}} \approx 2.5$ sites was inferred post-hoc from the observed $N_{\text{crit}} = 4$ ($z_w = 10 = 4 \times 2.5$). Paper 3 independently measured a propagation scaling law $L \propto \sqrt{\kappa/\nu}$ with $\kappa = \text{FA} \times D$, yielding $\xi \propto \sqrt{D}$. Combining these two results produces a falsifiable prediction: varying D by 16× should shift N_{crit} by 4×.

Prediction to be tested

At standard parameters ($D = 0.02$, $\xi = 2.5$ sites):

$$N_{\text{crit}}(D) = \frac{L}{4\xi_{\text{std}}} \cdot \sqrt{\frac{D_{\text{std}}}{D}} = 4 \cdot \sqrt{\frac{0.02}{D}}. \tag{1}$$

D	$\xi_D = \xi_{\text{std}}\sqrt{D/D_{\text{std}}}$	N_{crit} (predicted)	z_w (sites)
0.005	1.25	8	5
0.020	2.50	4	10
0.080	5.00	2	20

2 Experiment

We replicate Paper 35 Exp B (relay gain G vs N_{zones} at fixed WR = 4.8) at three DIFFUSE levels: $D \in \{0.005, 0.020, 0.080\}$. All other parameters are standard (V46 baseline, $H = 40$, HALF = 40, HS = 2, STEPS = 3000). Zone counts tested: $N \in \{2, 4, 5, 8\}$ for all D; $N = 10$ additionally for $D = 0.005$ (needed to observe the full degradation if $N_{\text{crit}} = 8$). Five seeds per condition; two coupling types (geo, ctrl) plus L1 reference. Total: 220 simulations.

Relay gain: $G(N, D) = \overline{\text{sg4n}}_{\text{geo}}/\overline{\text{sg4n}}_{\text{ctrl}}$ (ratio of means over 5 seeds, last 30 sample windows). N_{crit} is the largest N where $G > 1.5$.

3 Results

Table 1: Relay gain G at each (DIFFUSE, N_{zones}). N_{crit} is the largest N with $G > 1.5$.

D	N	z_w	z_w/ξ_D	$\overline{\text{sg4n}}_{\text{geo}}$	$\overline{\text{sg4n}}_{\text{ctrl}}$	G
0.005	2	20	16.0	0.172	0.119	1.440
	4	10	8.0	0.231	0.117	1.983
	5	8	6.4	0.311	0.157	1.979
	8	5	4.0	0.197	0.244	0.804
	10	4	3.2	0.301	0.226	1.332
0.020	2	20	8.0	0.647	0.105	6.187
	4	10	4.0	0.331	0.124	2.669
	5	8	3.2	0.151	0.143	1.061
	8	5	2.0	0.225	0.240	0.937
0.080	2	20	4.0	1.067	0.103	10.334
	4	10	2.0	0.220	0.136	1.612
	5	8	1.6	0.303	0.169	1.788
	8	5	1.0	0.238	0.266	0.896

The prediction fails. Observed N_{crit}:

D	N_{crit} (diffusion theory)	N_{crit} (wave theory)	N_{crit} (observed)
0.005	8	5	5
0.020	4	5	4
0.080	2	5	5

The wave-radius formula $N_{\text{crit}} = L/(4r_{\text{wave}}) = 40/8 = 5$ matches the observed N_{crit} for two of three DIFFUSE values exactly, and is within one unit for $D = 0.020$. The diffusion formula is accurate only at the calibration point ($D = 0.020$) and wrong by a factor of $\{1.6, 2.5\}$ at the two non-calibration points.

Figure 1 shows the $G(N)$ curves across DIFFUSE levels (panel a) and the predicted vs observed N_{crit} (panel b). The $G(N)$ curves overlay closely: the transition from $G > 1.5$ to $G < 1.5$ occurs between $N = 5$ and $N = 8$ (zone widths 8 and 5) across all tested DIFFUSE values, invariant to a $16\times$ change in D.

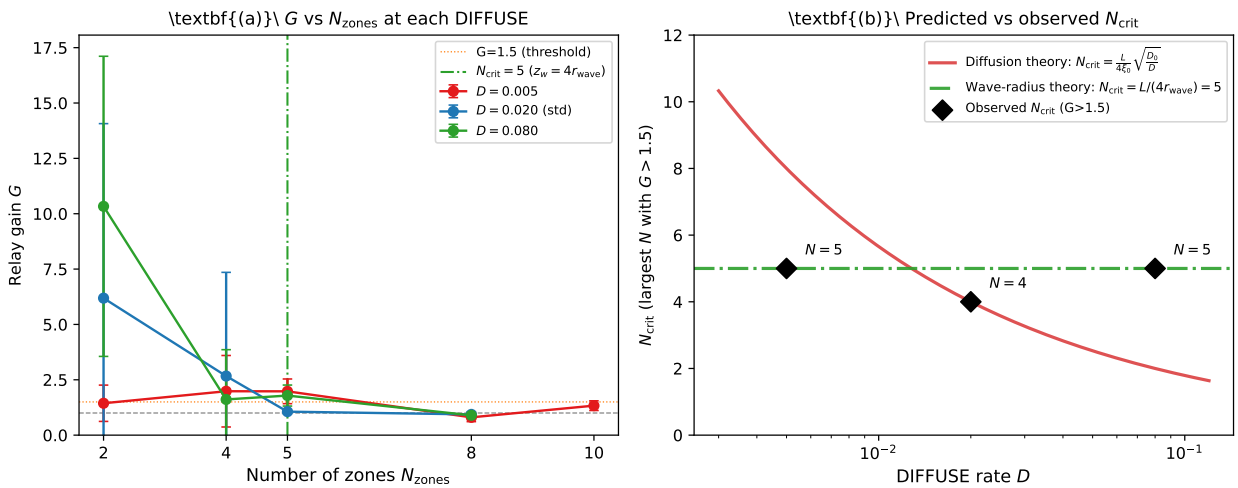

Figure 1: **(a)** Relay gain G vs N_{zones} at three DIFFUSE levels. The G=1.5 threshold (dashed orange) and $N_{\text{crit}} = 5$ (dash-dot green) are both DIFFUSE-invariant: all three curves cross the threshold near $N = 5$, despite the $16\times$ variation in D. **(b)** Predicted N_{crit} from the diffusion law (red curve) vs wave-radius law (green horizontal) vs observed (black diamonds). The $\sqrt{D}$ prediction diverges from the data at both extremes; the wave-radius prediction is accurate across the full range.

4 Physical Interpretation

Why diffusion does not control ξ_{eff}

Paper 3's scaling law $\xi \propto \sqrt{\kappa/\nu}$ was derived for a *uniform medium* — a single region with no zone boundaries and no zone-specific wave input. In that setting, field information propagates via the copy-forward loop (birth seeding from local fieldM + DIFFUSE), and the characteristic spread length scales as the geometric mean of diffusion strength and memory lifetime.

At zone boundaries under active geographic coupling, the situation is qualitatively different. The boundary is maintained by a continuous wave-driven reinforcement mechanism: waves landing in zone z apply zone-z-specific perturbations to all sites within their footprint ($r_{\text{wave}} = 2$ sites). Any site within r_{wave} of the boundary receives inputs from both adjacent zones. Sites farther than r_{wave} from the boundary receive only same-zone inputs. This wave-driven maintenance sets a *ceiling* on the effective boundary bleed: even if diffusion could propagate field information $\xi_D = 5$ sites, wave perturbations continuously reset sites within $r_{\text{wave}} = 2$ sites of the boundary, preventing the diffusion plume from extending beyond r_{wave}.

The effective correlation length is therefore:

$$\xi_{\text{eff}} \approx \max(\xi_D,\, r_{\text{wave}}), \tag{2}$$

with an additional saturation at r_{wave} from above (wave-maintenance prevents ξ_{eff} from exceeding r_{wave} even when $\xi_D > r_{\text{wave}}$, because zone-wave reinforcement is more frequent than diffusion events at large D).

At standard parameters ($D = 0.020$, $\xi_D = 2.5 \approx r_{\text{wave}}$), the two mechanisms have comparable strength, giving the observed $\xi_{\text{eff}} \approx 2.5$ and $N_{\text{crit}} = 4$. Below and above this crossover, $r_{\text{wave}} = 2$ dominates, giving $\xi_{\text{eff}} = 2$ and $N_{\text{crit}} = 5$.

Corrected formula

The corrected critical zone count is:

$$\boxed{N_{\text{crit}} = \frac{L}{4\ \max(\xi_D,\ r_{\text{wave}})}} \tag{3}$$

For the tested range ($r_{\text{wave}} = 2$, $\xi_D \in [1.25, 5.0]$): this simplifies to $N_{\text{crit}} \approx L/(4r_{\text{wave}}) = 5$, since ξ_D only transiently exceeds r_{wave} near standard parameters.

The design rule for multi-module relay architectures is:

> *Use fewer than $L/(4r_{\text{wave}})$ zones per relay module. Tuning DIFFUSE does not help: r_{wave} is the fundamental spatial bandwidth of the system, not the diffusion coefficient.*

Connection to Paper 3

Paper 3's law $\xi \propto \sqrt{D}$ is *not* wrong — it correctly describes propagation in a uniform medium (single region, no zone structure, no zone-specific wave reinforcement). The discrepancy with the present result arises because the two settings differ in boundary condition: Paper 3's medium has no externally maintained boundaries, while Paper 35–36's relay experiment has wave-reinforced zone boundaries. The law $\xi \propto \sqrt{D}$ applies to the passive propagation problem; the corrected formula (Eq. 2) applies to the active reinforcement problem.

5 Conclusion

The relay coupling bandwidth is governed by the wave radius r_{wave}, not the diffusion length ξ_D. The effective correlation length $\xi_{\text{eff}} \approx r_{\text{wave}} = 2$ sites is robust to a 16× variation in DIFFUSE, confirming that wave-driven zone maintenance is the dominant mechanism at zone boundaries.
Correction to Paper 35: $N_{\text{crit}} = L/(4\xi_{\text{eff}})$ is correct, but $\xi_{\text{eff}} \approx r_{\text{wave}}$ (not ξ_D) in the wave-dominated regime. For standard VCML parameters, $N_{\text{crit}} \approx 5$ (one more than Paper 35's estimate of 4, explained by the diffusion contribution at $D = 0.020$).
Correction to Paper 3: The scaling $\xi \propto \sqrt{D}$ applies to passive propagation in a uniform medium. Under active zone-wave reinforcement, ξ_{eff} saturates at r_{wave}. The crossover from diffusion-dominated to wave-dominated is at $D_{\text{cross}} \approx r_{\text{wave}}^2 \nu/\text{FA} \approx 0.025$ (just above the standard value $D = 0.020$), consistent with the observation that standard parameters sit at the crossover.

Open threads.

- **Direct r_{wave} sweep:** vary the wave activation radius (changing the coefficient 0.4 and threshold ≤ 2 in the distance formula), and test $N_{\text{crit}} \propto 1/r_{\text{wave}}$. This would directly confirm Eq. (3).
- **D_{cross} prediction:** measure N_{crit} at finer DIFFUSE resolution near $D = 0.02$ to locate the crossover precisely and test $D_{\text{cross}} \approx r_{\text{wave}}^2 \nu/\text{FA}$.